

License Plate Detection

Session 3

1. GOAL

- Analyse YOLO performance metrics
- Explore the output of a pre-trained YOLO
- Understand impact in performance of data set nature (domain shift)

2. MATERIALS:

Slides: Session slides 3.Challenge1_Detection_DL_Approach.pdf

Code: yolo_v1.py, LP-detection.pt, yolov8n.pt, yolov5nu.pt

Data: real_data.zip, your own data

3. SESSION EXERCICES

Exercise 1. Performance of YOLO trained on COCO Dataset. Evaluate YOLO8 on the set real_data.zip using the script yolo_v1.py.

- a) Study the YOLO pipeline: input, feature extraction, bounding box prediction, confidence scores, and Non-Maximum Suppression (NMS)
- b) Explore the distribution of the objects detected and its confidence score using boxplots and histograms.
- c) Visualize the detected regions
- d) Does the model accurately detect all cars in our data set or there is room for improvement? Evaluate the model on COCO images with cars, Why do you think the model does perform equally on our data?

Exercise 2. Performance of YOLO trained on specific licence plate images. Repeat the analysis of Exercice1 using the model LP-detection.pt

Exercise 3. YOLO fine tuning. Re-train LP-detection.pt, yolov8n.pt to adapt the models to our data set. Use the output of the ML detector to define a GT.

- a) Use the output of the ML detector to define a GT.
- b) Compare the performance of fine tuned YOLO model and the ML approach under varying acquisition conditions using different detection metrics